

Literature Review: Livelihoods, Vulnerability & Labour Markets

Kedarnath / Char Dham Ropeway Livelihood Assessment

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Roadmap: Five Strands of Literature

- ① Sustainable Livelihoods & Capitals Frameworks
- ② Vulnerability-to-Poverty (ex-ante risk)
- ③ Risk-Sharing, Insurance & Consumption Smoothing
- ④ Labour Market Dynamics
- ⑤ Infrastructure Impacts

Each strand closes with the gap it leaves for our study.

Strand 1: Sustainable Livelihoods Approach (SLA)

- **Chambers & Conway (1991)** – the landmark paper. A livelihood = capabilities + assets (stores, resources, claims, access) + activities; sustainable if it can “*cope with and recover from stress and shocks.*” No variables – purely conceptual. Ignores inequality of life across livelihoods, and doesn’t push the reader to look at interactions *within/between* communities – other authors call this “political capital,” worth exploring for us. Key aspects to evaluate per the author: stores/resources (e.g. cash savings), intergenerational mobility, access to resources (not just what people have – do mule/pony owners have equal access to grazing land?), and inequality across livelihoods (a Gini approach once we have a bigger sample).
- **Bebbington (1999)** – gives us the “Five Capitals” (Produced, Human, Natural, Social, Cultural), later adopted by DFID. Framework only, no variables.
- **Naithani & Saha (Kedarnath, 2013 disaster)** – households shifted from pilgrimage-based to skill-based livelihoods after the disaster; geographical isolation shaped the recovery trajectory via social capital (bridging/bonding/linking networks). Mixed-methods (change detection + secondary data + two-village case study). This is basically our own setting’s precedent.

Strand 1: Composite Livelihood/Security Indices

- **Mondal, Ghosh & Rohatgi** – village-level Livelihood Security Index via AHP (West Bengal). Dimensions/variables: natural resources, housing, drinking water & sanitation, economic condition, socio-cultural condition, assets. Explicitly splits **sensitivity to stress/shocks** from **capacity to cope** – a good way to structure our own shocks section.
- **Singh & Hiremath** – SLSI = Ecological Security Index + Economic Efficiency Index + Social Equity Index (Gujarat, district level). Variables are a long catalogue: GDP, PQLI, HDI, NPP, Ecological Footprint, Environmental Sustainability/Vulnerability Index, PSR model, Well-Being Index, Genuine Savings Index, etc. Too macro to use directly – question for us is which of these we can actually enumerate for our own population (PQLI, HDI, ecological footprint for mule owners, environmental vulnerability from terrain data, PSR, well-being, genuine savings).
- **Patidar** – national/state-level livelihood security index, HDI-style weighting of 23 variables across six domain indices. Documents big regional gaps (Chandigarh/Goa/Kerala high, Bihar/Jharkhand/Chhattisgarh low).

Gap this strand leaves

None of these go down to the household level with primary data – they're all secondary-data indices at village/district/state level.

Strand 2: Vulnerability-to-Poverty (Ex-Ante Risk)

- **Chaudhuri, Jalan & Suryahadi (Indonesia)** – the foundational method: estimate $\ln c_h = X_h\beta + e_h$, model the mean *and* variance of log consumption via FGLS, assume log-normality, then compute each household's probability of falling below the poverty line – from a **single cross-section**, no panel needed. Vulnerability (45%) is much higher than observed poverty (22–23%). Variables: household characteristics X_h feeding the consumption equation. This is the key takeaway for Kedarnath – you don't need panel data to build a forward-looking vulnerability measure, which matters since a household panel is probably infeasible for us.
- **Khosla, Jena & Rahut (tribal Odisha)** – applies the Chaudhuri method (FGLS → multinomial logit for chronic/transient poverty → Endogenous Switching Regression) to a livelihood program (OTELP). VtP (34.7%) > actual poverty (28.6%); the program cuts VtP by ~3pp (ATT). **Most relevant paper for us so far** – full variable list: gender, farm-employed, wage in non-farm, household size, dependency ratio, age of head, years of schooling of head, own land, durable assets, illness, death of breadwinner, cyclone/flood/drought, sold livestock/land/gold, borrowed from moneylender/relatives, productive assets, SHG/saving-group membership, attending public meetings. Also builds a resilience/deprivation index across education, health and standard-of-living sub-indicators (schooling, school attendance, chronic illness, child mortality, housing, cooking fuel, drinking water, electricity, sanitation, asset ownership) – basically a ready-made checklist we can adapt.

Strand 3: Risk-Sharing & Insurance (1/2)

- **Jalan & Ravallion (rural China)** – GMM treating income and household size as endogenous; efficient risk-sharing model. Full insurance is rejected everywhere, but the poorest decile still passes 40% of an income shock into current consumption vs. $\sim 10\%$ for the richest third. Food consumption is better protected. Variables: household consumption (food vs. total), income, household size. Income risk is pervasive and household behaviour is geared around protecting consumption from it – directly relevant since our workers live off 6 months of income. Open question we noted: is low credit-taking by vulnerable households actually a sign of credit/risk-market failure?
- **De Weerd & Dercon (Tanzania)** – risk-sharing happens through **overlapping informal networks**, not the whole village. Elicited with one question: “list everyone you depend on for help.” Also gives us a shock taxonomy with % of households forced to cut consumption per shock type: death/funeral, ceremonies, sickness, lumpy expenditure (house-building, education), crime/court cases, income-generating-activity shocks. This is the literature that fills the “missing political capital” gap from Chambers & Conway – we can use the same “who would you turn to” question and segment it by occupation group.

Strand 3: Risk-Sharing & Insurance (2/2)

- **Islam & Maitra (Bangladesh, panel, 4 waves)** – regression of consumption differentials on household controls + health-shock dummy, with village/year FE and interactions; random-effects Tobit for censored borrowing. Households sell livestock to smooth consumption after a health shock; microcredit access substitutes for asset sales – an insurance role for microfinance. Their own suggested health module (worth exploring): number of sick days in the Yatra season, number of days refrained from work due to sickness – caveat that self-reported sickness is culturally conditioned.

Strand 4: Labour Market Dynamics

- **Abdulla, Amir & Khan (India, pooled PLFS 2017–18 to 2023–24)** – logistic regressions for informal-work participation and social-security receipt, log-linear regression for income, Blinder-Oaxaca decomposition of the formal–informal wage gap, PCA to build a livelihood index. Education is the biggest predictor across the board (raises income, lowers informal participation, raises social-security access). Income penalty from BO decomposition is bigger for rural and informal workers – reflects structural segmentation, not just differences in endowments. **Vocational training is not significant in any specification – a training-to-working mismatch.** Variables: informal-employment status, social-security receipt, individual earnings, household consumption, education, vocational training, state & survey-year FE. Directly informs what to control for in our own income/informality regressions, and warns us not to over-trust “willingness to retrain” as a proxy for actual mobility.

Strand 5: Regional/Infrastructure Impact & Method (1/2)

- **Datta** – Golden Quadrilateral highways, diff-in-diff on World Bank Enterprise Surveys (2002 & 2005). Variables: treatment = on/off the upgraded highway + distance from it; outcomes = input-inventory days held (–6 to –12 days on-highway), supplier switching, self-reported transport obstacles. This is our template for a quasi-experimental design – villages connected vs. not-yet-connected by the ropeway route are “as good as random” at the margin, similar to an RDD across a fixed threshold.
- **Chhetri, Chhetri, Arrowsmith & Corcoran (Australia)** – Location Quotient, Moran's I, Spatial Error Model on employment data. Finds positive spatial clustering of tourism/hospitality employment. Tells us we're likely to see serial correlation in our own survey responses driven by space, and we should instrument for / control it to defend our inferences.

Strand 5: Regional/Infrastructure Impact & Method (2/2)

- **Hussain et al. (Pakistan)** – PLS-SEM, Cronbach's alpha, CR, AVE on a 213-respondent survey of construction professionals. Economic-factor and social-factor composites both significantly drive community development ($\beta = 0.499$, $t = 12.022$ for social factors). Notes: this is trial-and-error path-coefficient work and probably not our best approach.
- **Chan et al.** – CFA on a Likert-survey + focus groups. Validates a long list of infrastructure-impact indicators (project management, materials, ecology, water, waste, energy, community impact – noise/vibration/light/traffic, stakeholder engagement, health & safety, heritage) into broader dimensions. Could be useful to test whether our own locally-identified ropeway impacts cluster into economic/social/environmental/accessibility dimensions – but factor loadings shouldn't be read as normative importance weights.

Cross-Cutting Gaps Across All Five Strands

- 1 SLA frameworks (Strand 1) are almost never run at household level with primary data, and never for a seasonal, single-industry pilgrimage economy.
- 2 VTP methods (Strand 2) are transferable but untested in a setting where nearly all income lands in one 6-month window.
- 3 Risk-sharing papers (Strand 3) put informal networks front and centre, but SLA frameworks skip “political/network capital” entirely – no paper combines the two.
- 4 Labour literature (Strand 4) documents the informality wage penalty but never asks whether the labour market can actually *absorb* a shock – no tightness or matching-friction measure exists for informal, route-based work like ours.
- 5 No paper combines an ex-ante vulnerability index with a labour-market recovery/mobility lens for a single, shock-exposed, tourism-dependent occupation cluster.

Further Reading Needed

Vulnerability-to-Poverty / multidimensional poverty

We need to explore Alkire & Foster's original multidimensional poverty methodology directly, rather than relying on how Fahad et al. and Khosla et al. used it second-hand. Before we lock in dimensions, cutoffs and weights for our own variables, we need to go through that ourselves.

Labour market dynamics

This strand is currently thin – only one applied paper (Abdulla et al.) reviewed so far. We need to explore frictional/matching labour-market models in more depth, and how labour market tightness is actually measured in the literature, before we build out the “labour market tightness / can the market recover from a shock” angle.

Proposal 1: Generate Good Variables Before Any Index

This is our synthesis, not literature. Almost every paper in Strand 1 and Strand 2 (SLSI, LSI, AHP livelihood security, Fahad et al.'s deprivation index, Khosla et al.'s resilience index) jumps straight to a composite index with pre-assigned weights, without first testing which variables actually matter for their population. We want to avoid that.

- **Pool the variable catalogues first:** capitals variables from Strand 1 (Mondal et al./Singh & Hiremath/Patidar), VTP covariates from Strand 2 (Khosla et al.'s full variable list), network/shock variables from Strand 3 (De Weerd & Dercon's shock taxonomy, Islam & Maitra's health module).
- **Test them properly:** run these variables through a determinants model (FGLS/logit, following Chaudhuri et al. / Khosla et al.) on our own data, rather than assigning equal weights upfront.
- **Index construction, if any, comes later:** only once we know which variables/dimensions actually carry signal for our population – not before.

Proposal 2: A Labour-Market Lens on the Research Question

Reframe: instead of only asking how much income a household loses, ask whether the local labour market can actually absorb the shock.

- **Labour market tightness:** ratio of available Yatra-linked work/route-permits to workers seeking it; queueing for dispatch/routes; unmet demand for handlers/porters at peak.
- **Entry/exit & reallocation:** new entrants vs. incumbents per season; whether entry is gated by union/permit systems.
- **Shock recoverability:** use the 2013 Kedarnath disaster and COVID closures as real pre/post shocks – how long until respondents re-entered work, and did they re-enter the *same* occupation? (Naithani & Saha already show occupations shifted post-2013 – we can build directly on that.)
- **Realised mobility, not stated willingness:** Abdulla et al. already warn that stated willingness to retrain is uninformative if the underlying work is structurally changing – so we should measure *past* occupational transitions, not just intent to shift.

We need to explore this further

What We Already Ask

We already cover: household & occupation profile, income sources, work economics for shops/stalls, productive assets, debt & savings, shocks (climate/disaster only), ropeway awareness, expected livelihood impact, skills & willingness to shift occupation, land/natural capital, and attitude statements.

The five gaps below are where the literature review most directly changes what we should be asking.

Gap 1: Our Shocks Section Only Covers Climate/Disaster Events

Gap

We currently only ask about covariate shocks – cloudburst, landslide, flood, road blockage, snowfall, pandemic. De Weerd & Dercon show idiosyncratic shocks (death/funeral, sickness, lumpy expenditure, crime) are just as consequential for consumption cuts.

What we plan to add: household-level idiosyncratic shocks (major illness, consumptions shocks) alongside the existing climate list, plus a recovery-time follow-up – how many weeks/months before you resumed normal work/income after the most serious shock? This is our direct handle on “can the labour market recover from even basic shocks.”

Gap 2: No Network / “Political Capital” Questions

Gap

We only capture formal union/cooperative membership and the Pilot reveals scarce data for this measure. It might be worth capturing informal risk-sharing networks, which De Weerd & Dercon show matter more than village-level insurance.

What we plan to add: “In the event of an income or health shock, who would you turn to for help?” (household/relatives, occupation peers, union, moneylender, none) – and the mirror question, who turns to you. Lets us map informal insurance networks by occupation group, filling the “missing political capital” gap from Chambers & Conway.

Gap 3: No Consumption/Expenditure Data

Gap

We capture income sources and months of food shortage, but nothing on expenditure. The Chaudhuri–Jalan–Suryahadi vulnerability method (Proposal 1) needs a consumption measure, and Jalan & Ravallion's insurance test is run on consumption, not income.

What we plan to add: monthly food and non-food expenditure (coarse buckets are fine) – enables both the ex-ante vulnerability model and a direct consumption-smoothing/insurance test.

Gap 4: No Standard Deprivation Indicators

Gap

To build the variable set proposed in Proposal 1, following the indicator lists Fahad et al. and Khosla et al. used, we need standard deprivation indicators. Right now we only capture house type and land.

What we plan to add (reusing the Khosla/Fahad indicator sets already in our notes): drinking-water source, sanitation/latrine access, cooking-fuel type, electricity access, durable-asset ownership count, child school attendance.

Gap 5: We Ask Willingness to Shift, Not Actual Mobility or Tightness

Gap

We measure *stated* willingness to retrain/migrate, but not realised past occupational transitions, entry barriers, or market tightness – and our own notes on Abdulla et al. warn that stated willingness is uninformative without structural mobility.

What we plan to add:

- Structured occupational history – number of past occupation switches, and whether each was shock-driven or opportunity-driven.
- Entry friction – how did you get your current route/permit/position (inherited, purchased, union allocation, open entry)? Was there a queue/waitlist?
- Tightness proxy – at peak season, did you turn away work/customers, or go without work, last season?
- Pre/post-2013 and pre/post-COVID re-entry – did you work in this occupation before 2013 / before 2020? If work stopped, how long until you resumed?

Summary

- Five strands reviewed: SLA/capitals, vulnerability-to-poverty, risk-sharing/insurance, labour market dynamics, regional-impact/identification methods.
- Two areas flagged for deeper reading before we commit: multidimensional poverty measurement (Alkire-Foster) and labour-market tightness/frictional models.
- Proposed synthesis: a household-level, multidimensional variable set for ex-ante vulnerability (Proposal 1) read jointly with a labour-market tightness and shock-recovery lens (Proposal 2).
- Five concrete, literature-motivated questionnaire gaps: shock typology too narrow, no network/insurance questions, no consumption data, no deprivation indicators, no realised-mobility/tightness questions.

Discussion / edits